

RQ Answers

omitted

2024-04-05

Data overview

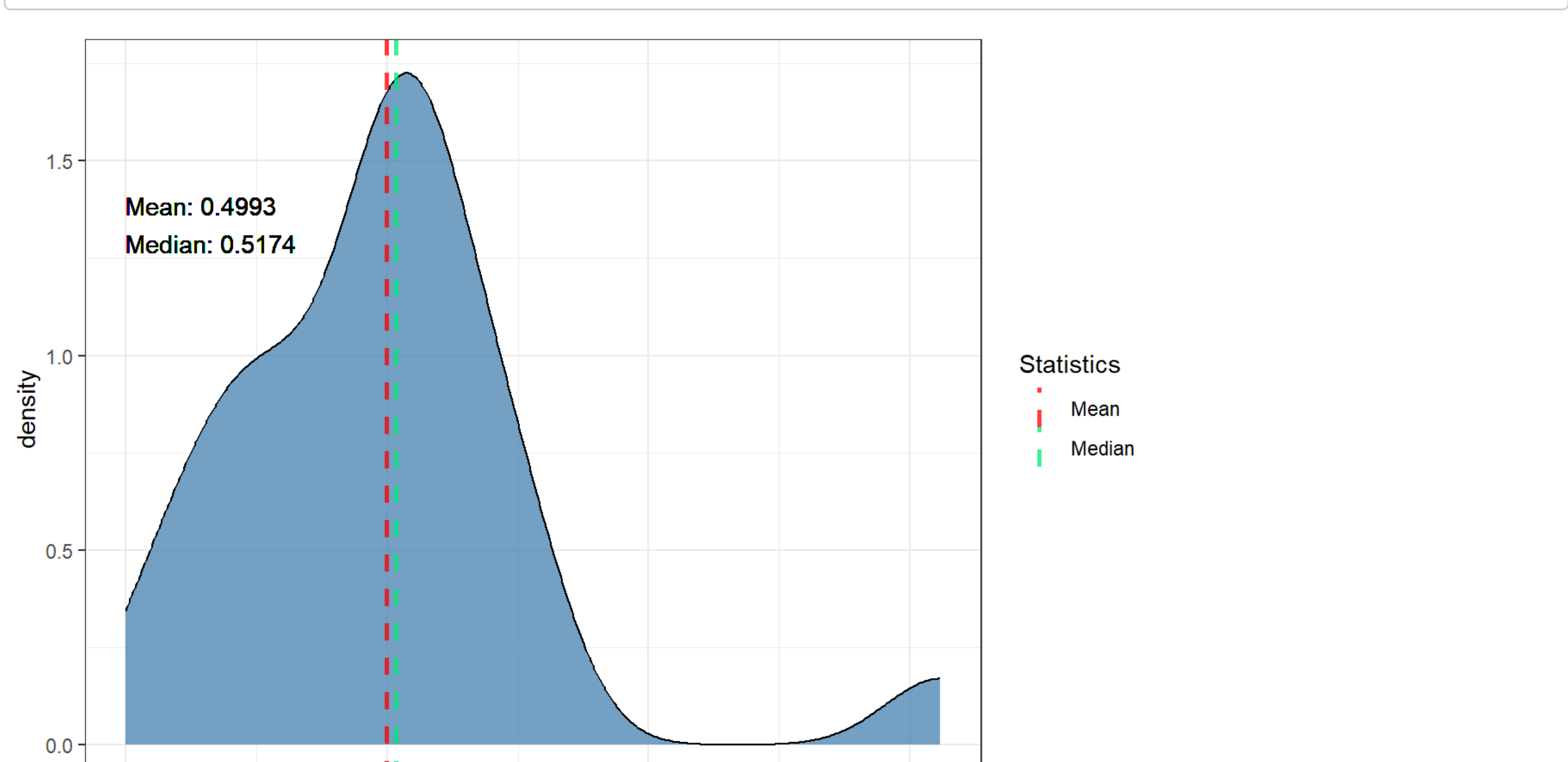
glimpse(all_data)				
## Rows: 1,149				
## Columns: 11				
## \$ repo	<chr>	"zookeeper", "zookeeper", "zookeeper", "zookeeper"...		
## \$ pr_number	<int>	1697, 1697, 1799, 1799, 1799, 1799, 1799, 1799, 17...		
## \$ rule	<chr>	"java:S1192", "java:S125", "java:S1075", "java:S10...		
## \$ severity	<chr>	"CRITICAL", "MAJOR", "MINOR", "MAJOR", "CRITICAL",...		
## \$ file	<chr>	"zookeeper-server/src/test/java/org/apache/zookeep...		
## \$ type	<chr>	"CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...		
## \$ status	<chr>	"OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...		
## \$ debt	<int>	8, 5, 20, 5, 8, 5, 5, 5, 34, 20, 5, 15, 15, 5, 10,...		
## \$ ncloc_affected_file	<int>	64, 64, 340, 340, 340, 340, 340, 340, 340, 340, 34...		
## \$ complexity	<int>	5, 5, 46, 46, 46, 46, 46, 46, 46, 46, 46, ...		
## \$ origin	<chr>	"PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...		

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 22 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 22 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```

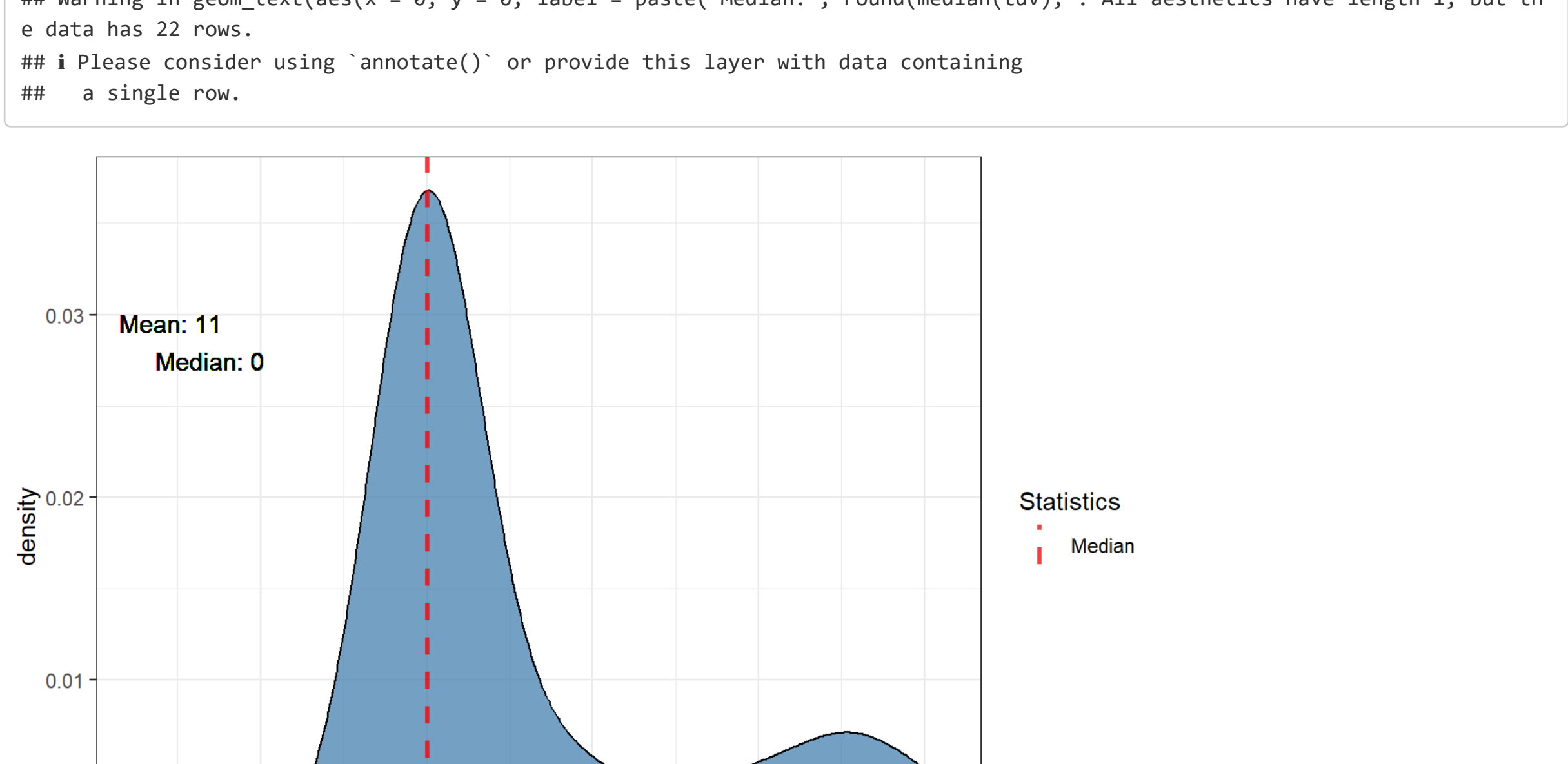


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 22 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 22 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 zookeeper	9.09	54.6	36.4	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	<dbl>
## 1	9.09	54.6	36.4

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 zookeeper	77.3	22.7	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	77.3	22.7

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 zookeeper	80	20	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	80	20

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank zookeeper		
## <int> <chr>		
## 1	1 java:S100	- 12
## 2	2 java:S112	- 3
## 3	3 java:S1192	- 3
## 4	4 java:S3776	- 2
## 5	5 java:S2184	- 2
## 6	6 java:S2093	- 2
## 7	7 java:S5993	- 1
## 8	8 java:S1161	- 1
## 9	9 java:S4042	- 1
## 10	10 java:S899	- 1

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S100	122	
## 2 java:S112	123	
## 3 java:S1192	124	
## 4 java:S2093	125	
## 5 java:S2184	126	
## 6 java:S3776	127	
## 7 java:S1161	128	
## 8 java:S3011	129	
## 9 java:S4042	130	
## 10 java:S5993	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank zookeeper		
## <int> <chr>		
## 1	1 java:S1192	- 168
## 2	2 java:S2095	- 91
## 3	3 java:S2142	- 74
## 4	4 java:S899	- 63
## 5	5 java:S2184	- 53
## 6	6 java:S1135	- 52
## 7	7 java:S4042	- 44
## 8	8 java:S108	- 37
## 9	9 java:S6213	- 37
## 10	10 java:S1075	- 34

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1192	122	
## 2 java:S2095	123	
## 3 java:S2142	124	
## 4 java:S899	125	
## 5 java:S2184	126	
## 6 java:S1135	127	
## 7 java:S4042	128	
## 8 java:S108	129	
## 9 java:S6213	130	
## 10 java:S1075	131	

Outliers

Higher TDD

## # A tibble: 22 × 5				
## repo	pr_number	total_debt	total_ncloc	td
## <chr>	<int>	<int>	<int>	<dbl>
## 1 zookeeper	2128	363	233	1.56
## 2 zookeeper	1915	99	131	0.756
## 3 zookeeper	2001	356	483	0.737
## 4 zookeeper	2025	154	210	0.733
## 5 zookeeper	2100	3143	4966	0.633
## 6 zookeeper	2099	78	130	0.6
## 7 zookeeper	2134	2328	4130	0.564
## 8 zookeeper	2008	975	1732	0.563
## 9 zookeeper	1942	1967	3632	0.542
## 10 zookeeper	2115	344	637	0.540
## # i 12 more rows				

Higher TDV

## # A tibble: 22 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	<int>
## 1	2115 zookeeper	62	
## 2	2128 zookeeper	53	
## 3	2064 zookeeper	50	
## 4	2065 zookeeper	42	
## 5	2001 zookeeper	33	
## 6	2093 zookeeper	20	
## 7	2087 zookeeper	15	
## 8	2067 zookeeper	5	
## 9	1697 zookeeper	0	
## 10	1799 zookeeper	0	
## # i 12 more rows			

Lower TDV

## # A tibble: 22 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	<int>
## 1	2100 zookeeper	-35	
## 2	2025 zookeeper	-3	
## 3	1697 zookeeper	0	
## 4	1799 zookeeper	0	
## 5	1915 zookeeper	0	
## 6	1942 zookeeper	0	
## 7	1944 zookeeper	0	
## 8	1985 zookeeper	0	
## 9	1992 zookeeper	0	
## 10	2008 zookeeper	0	
## # i 12 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 22	

Number of java:S1192 issues that affect test code

## n	
## 1 171	

Number of java:S1192 issues that affect not test code

## n	
## 1 0	

Summary NCLOC

## ncloc	
## Min. :	64.0
## 1st Qu.:	138.0
## Median :	335.0
## Mean :	916.1
## 3rd Qu.:	626.2
## Max. :	4845.0